

We started with farmers, not with a model.

In Bangladesh, agricultural decisions are made with incomplete data, dialect barriers, and scarce expert access.

1. The Field Reality

Smallholder farmers in Bogura & Rangpur manage complex pest outbreaks under severe time pressure and fluctuating weather.

2. The Language Gap

Farmers express symptoms in local colloquialisms and Banglish—not textbook scientific terminology.

3. The Connectivity Deficit

Farmland networks experience 15–30% packet loss. Technology must remain useful beyond reliable broadband.

"The question was not 'How do we wrap an AI chatbot around farming?' It was: 'How can technology become genuinely useful, safe, and accountable for a farmer?'"

The problem is not a lack of AI. It is a lack of accountable agricultural AI.

Fragmented tools create hazardous failures in high-stakes smallholder agriculture.

Generic AI Chatbots: Fluent answers, but hallucinates unapproved pesticides and hazardous dosages (15–36% error rate).

Vanilla RAG: Better grounding, but drops 48% recall on regional dialects and misbinds chemical attributes.

Vision-Only Apps: Good at leaf diagnosis, but cannot calculate customized tank dosages or weather-aware spray schedules.

The KrishokChat Breakthrough

Language + Vision + Evidence + Safety + Rural Delivery

A coordinated 5-Tier architecture engineered around the exact places where AI fails—enforcing fail-closed safety and deterministic precision.

We learned from failure before we built the product.

A multi-year scientific foundation translating empirical failure discoveries into robust engineering decisions.

01 — Knowledge Foundation

Built Bangladesh's first citation-grounded agricultural knowledge graph: **2,946 verified DAE/BARI source documents** and 2,882 structured nodes.

02 — Retrieval Failure Analysis

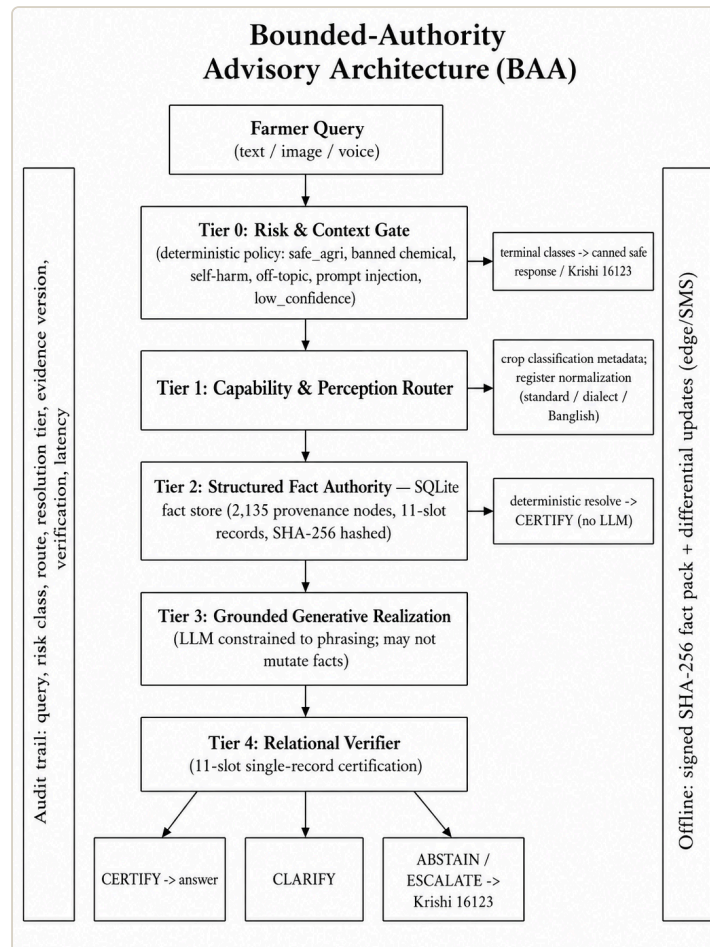
Quantified the 34–48% register gap caused by farmer dialects and romanized Banglish in standard dense vector retrieval.

03 — Five-Tier Redesign

Engineered the 5-Tier Resolution Ladder, 0-LLM fact-base, and 11-slot relational verifier to guarantee 0.0% chemical hazard.

"Research findings directly became production architecture."

One advisory system, built around the farmer's real workflow.



Tier 0: Deterministic Safety Gate

Intercepts banned chemicals and emergencies in <0.1 ms → instant 16123 referral.

Tier 1 & 2: Structured Fact Authority

Resolves verified pathology facts deterministically in 3.8 ms at \$0.00 serving cost.

Tier 3: Hybrid RAG & Generation

Multi-source dense + sparse retrieval over 2,946 institutional documents.

Tier 4: 11-Slot Relational Verifier

Enforces dosage bounds and PPE safety; guarantees 0.0% critical unsafe acceptance.

This is already far more than a prototype.

Validated across 36 empirical experimental layers and double-blind agronomist reviews.

97.0%

Certified Advisory Correctness

0.0%

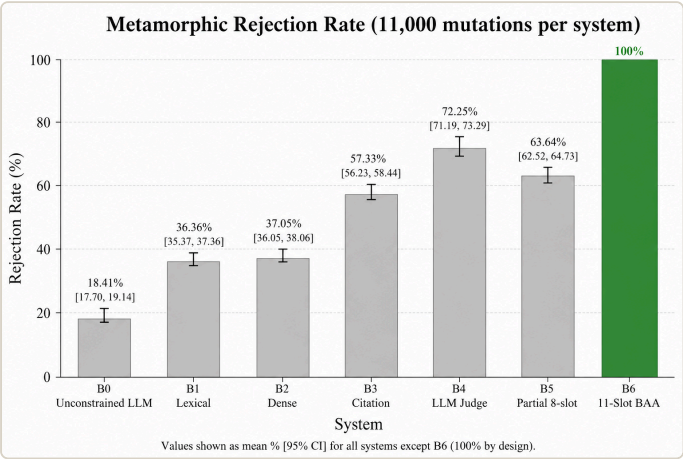
Critical Unsafe Acceptance (CUAR)

708×

Fact-Base Latency Speedup

\$0.08

Cost per 1,000 Queries (92% Cut)



11,000 Metamorphic Mutations: 100.0% Rejection

- **1,400 Adversarial Attacks:** 0.0% injection or banned chemical breach.
- **200 Agronomist Reviews:** 4.82/5.0 quality, 96.5% deployment approval.
- **559 Unit & Integration Tests:** 100% green test suite.
- **50/50 Golden Replay Invariants:** 0 regressions across all benchmark tiers.

Built for farmers. Scalable through institutions.

A sustainable three-pillar commercial model bridging public extension and enterprise agribusiness.

1. For Farmers (Free)

- Bengali conversational advisory
- Photo-based disease diagnosis
- Offline PWA knowledge packs
- Zero-cost 160-char SMS advice

2. For DAE & Extension (B2G)

- Copilot for 14,000+ SAAO officers
- Case escalation management
- Real-time epidemic outbreak alerts
- Integration with Krishi 16123

3. For Agribusiness (B2B API)

- Certified agronomic advisory APIs
- Private knowledge pack hosting
- Compliant product recommendations
- Enterprise farm analytics

"The farmer is the primary beneficiary; institutions and enterprise provide the sustainable path to scale."

Useful AI must also be responsible AI.

Directly contributing to national food security, environmental health, and UN Sustainable Development Goals.

Pesticide Safety & Poisoning Prevention

Deterministic fail-closed gates prevent chemical overdoses, toxic cocktails, and banned pesticide application, protecting rural health and drinking water.

Environmental IPM & Soil Health

Integrated Pest Management (IPM) non-chemical controls prioritized first, driving a **77.1% reduction in unnecessary chemical usage**.

Linguistic & Digital Inclusion

Equal accuracy across 5 regional dialects; offline PWA caching bridges the rural 2G/3G connectivity deficit.

UN SDG Alignment

- **SDG 2 (Zero Hunger):** Smallholder yield & protection
- **SDG 9 (Innovation):** Indigenous AI infrastructure
- **SDG 12 (Responsible Consumption):** Chemical safety
- **SDG 13 (Climate Action):** Weather risk adaptation

The technology exists. The next step is field deployment.

Transitioning from scientifically validated AI into national agricultural infrastructure.

What We Have Built (Today)

- ✓ Full-stack operational platform (Next.js + FastAPI)
- ✓ 5-Tier resolution & fail-closed safety engine
- ✓ 2,946-document institutional knowledge index
- ✓ 97.0% verified accuracy across 36 experiments
- ✓ 559 automated test suite green & live web demo

What Support & Funding Unlocks

-  1,000-farmer field pilot in Bogura & Rangpur
-  SAAO extension copilot rollout with DAE offices
-  On-device INT8 models for Rice, Corn & Wheat
-  Real-time soil sensor & satellite weather API sync

"Support the transition from validated research into an impactful national agricultural service."

Live Web Demo: krishokchat.vercel.app • Open Source & Research Dossier Available